

AMP Challenge 2027 — data disclosure

Hazra group submission entry.

1. Sources

No non-public data was used. All four sources are public.

source	obtained from	local path	dataset size
DBAASP	REST API, scraped by scripts/dbaasp/01-04	data/raw-data/dbaasp/	25,069 peptides
GRAMPA	zswitten/Antimicrobial-Peptides — data/grampa.csv	data/raw-data/grampa/grampa.csv	6760 peptides
DRAMP	https://dramp.cpu-bioinform.org/downloads/Mendeley	data/raw-data/dramp/	see below
MarlysAMP	10.17632/w4hb5grjwb.3	data/raw-data/marlys-amp/MLAMP_db.json	103,200 peptides

The raw responses scraped from dbaasp are zipped and provided here:

- data/raw-data/dbaasp/list_pages.zip — every paginated GET /peptides response
- data/raw-data/dbaasp/details.zip — every GET /peptides/{dbaaspId} record

DBAASP dataframes (built by pipeline):

step	file	rows
02	dbaasp-peptides-compiled.csv	25,069
04	dbaasp-activity.csv	193,763
04	dbaasp-hc.csv	34,378

GRAMPA raw data can be downloaded using the following commands:

```
mkdir data/raw-data/grampa
wget https://raw.githubusercontent.com/zswitten/Antimicrobial-Peptides/refs/heads/master/data/grampa.csv -O
data/raw-data/grampa/grampa.csv
```

DRAMP ships two files:

file	rows
Antibacterial_amps.txt	28,711
general_amps.txt	14,652

The raw DRAMP files are zipped and provided at: data/raw-data/dramp.zip.

Marlys AMP data can be downloaded like so:

```
mkdir data/raw-data/marlys-amp
wget
↪ https://data.mendeley.com/public-files/datasets/w4hb5grjwb/files/6c831932-
↪ 8e0f-4441-9e53-33fae57da262/file_downloaded -O
↪ data/raw-data/marlys-amp/MLAMP_db.json
```

2. Data pipeline — stages 1 and 2

To be run in the following order from repo root. Every script is argparse-driven with explicit -- paths; add suitable arguments whenever necessary.

2.1 Scrape (DBAASP only)

```
python scripts/dbaasp/01-fetch-list-pages.py           # paginated peptides
↪ information
python scripts/dbaasp/02-build-peptides-csv.py         # builds
↪ dbaasp-peptides-compiled.csv (25,069)
python scripts/dbaasp/03-fetch-peptide-details.py      # fetch detailed
↪ peptide records
python scripts/dbaasp/04-build-peptide-activity-hc-df.py # segregates
↪ peptides' data into two CSVs: activity (193,763) and hemolysis (34,378)
```

To reproduce the DBAASP dataframes used for training,

```
unzip /path/to/list_pages.zip                         # paginated peptides
↪ information
python scripts/dbaasp/02-build-peptides-csv.py         # combine and flatten
↪ peptide records
unzip /path/to/details.zip                            # fetch detailed
↪ peptide records
python scripts/dbaasp/04-build-peptide-activity-hc-df.py # segregates
↪ peptides' data into two CSVs: activity (193,763) and hemolysis (34,378)
```

2.2 Clean and combine

```
python scripts/clean_and_combine/prepare-finetune.py \
--dbaasp ... --grampa ... --dramp ... --out
↪ data/processed-data/clean_and_combine/finetune.csv --fasta
↪ data/processed-data/clean_and_combine/finetune.fasta # finetune.csv
↪ (8,455 peptides)
```

Filters three data sources to keep: - monomer only - 20 standard proteinogenic aa - 8-50 aa - no N/C-terminal modifications - no intra-chain bonds (e.g. disulfide).

Source DB of peptide sequences in finetune data

Source DB	sequences
DBAASP	6,521
GRAMPA	4,121
DRAMP	108

Only 108 sequences from DRAMP passed the above-mentioned filters.

```
python scripts/clean_and_combine/prepare-pretrain.py \
  --marlys ... --out data/processed-data/clean_and_combine/pretrain.csv
  ↪ --fasta data/processed-data/clean_and_combine/pretrain.fasta #
  ↪ pretrain.csv (34,937 peptides)
```

Filters the MarLys AMP data dump for length and disulfide only — Marlys AMP has no modification fields, so the terminal-modification filter cannot be applied. This is the larger set on which pre-training is carried out. `pretrain.csv` and `finetune.csv` have to be saved at the specified locations, since config YAML files use them.

2.3 Cluster and split

CD-HIT is run to cluster peptide sequences. Outputs are saved under `data/processed-data/cluster_and_seggregate/cd-hit-output/`.

CD-HIT version: V4.8.1 (+OpenMP), Aug 20 2021, 08:39:56

finetune command: `cd-hit -i finetune.fasta -o finetune50_output -c 0.5 -n 3 -l 5 -d 0 -M 0 -T 0` # out - finetune50_output.clstr 1,493 clusters

pretrain command: `cd-hit -i pretrain.fasta -o pretrain50_output -c 0.5 -n 3 -l 5 -d 0 -M 0 -T 0` # out - pretrain50_output.clstr 6,363 clusters

```
python scripts/cluster_and_seggregate/cdhit-split.py \
  --clstr data/processed-data/cluster_and_seggregate/cd-hit-
  ↪ output/finetune50_output.clstr --reference /path/to/finetune.csv
  ↪ \
  --val-frac 0.05 --test-frac 0.05 --out data/processed-
  ↪ data/cluster_and_seggregate/seggregate/finetune_seggregated.csv
```

Assigns the split at the cluster level, never splitting a cluster, using greedy bin-packing over cluster size and length bins (8-20 / 21-35 / 36-50 aa). Outputs 8,455 rows; train 7,609 / val 423 / test 423.

```
python scripts/cluster_and_seggregate/filter-pretrain.py \
  --cluster .../pretrain50_output.clstr --pretrain .../pretrain.csv \
  --finetune data/processed-
  ↪ data/cluster_and_seggregate/seggregate/finetune_seggregated.csv --out
  ↪ data/processed-
  ↪ data/cluster_and_seggregate/seggregate/pretrain_seggregated.csv
```

Drops pretrain peptides whose cluster overlaps the finetune val/test splits. Output: 28,121 peptides (from 34,937; 6,816 dropped).

Output paths for the segregated pretrain and finetune sets have to be at the specified locations since config YAML files use these locations.

2.4 Model inputs

```
python -m src.data.clean_corpus      # -> data/processed/*_clean.csv
python -m src.data.build_ld_labels   # -> data/ld-processed/*.csv
```

output	rows	columns
data/processed/finetune_clean.csv	8,455 (train 7,609 / val 423 / test 423)	id, sequence, split
data/processed/pretrain_clean.csv	28,091 (30 duplicates dropped)	id, sequence
data/ld-processed/finetuning.csv	8,455	sequence, charge, length, hydrophobic_moment, gravity
data/ld-processed/pretraining.csv	28,091	sequence, charge, length, hydrophobic_moment, gravity

- clean_corpus does a final check to ensure peptide sequences are valid according to competition conditions. Only change it made: 30 duplicates dropped from pretrain dataframe (28,121 -> 28,091).
- Both directories (data/processed/ and data/ld-processed/) are gitignored and rebuilt by the two commands above.

3. MIC label pipeline — stage 3

Separate from the corpus pipeline, and the data behind the regressor for ranked top-100.

```
python scripts/dbaasp/05-standardize-mic.py      # default --medium MHB,
↪ data/processed-data/dbaasp-std/filtered-dbaasp.csv
```

From dbaasp-activity.csv (193,763 rows), filters for - monomer only - measurement MIC only - regex-based special character cleaning of MIC values - $\mu\text{g/ml}$ $\rightarrow$ μM via molecular weight - censoring recorded before bound removed - target strains collapsed onto a species panel - 20 standard proteinogenic aa - 8-50 aa - no N/C-terminal modifications - no intra-chain bonds (e.g. disulfide)

Outputs data/processed-data/dbaasp-std/filtered-dbaasp.csv, with 2,344 unique sequences, all have medium MHB. Censored dataset: 2,660 right-censored peptides (MIC > mentioned concentration), 28 left-censored peptides (MIC < mentioned concentration).

```
python scripts/mic/01-build-mic-labels.py
```

Asserts the input is MHB-only, $y_{\text{pmic}} = 6 - \log_{10}(\mu\text{M})$ so higher is more potent. Censoring flips from MIC space into pMIC space due to sign change. Deduplication to one cell per (sequence, species), median of exact values or the weakest true bound, joins the cluster and split from the same CD-HIT run as stages 1-2, keeps the six best-supported species.

Output: data/processed-data/mic-labels/mic_labels.csv, 6,509 cells / 2,342 sequences.

Splits

split	count
train	2,099
val	133
test	110

Train clusters

	count
train clusters	495

Censoring (pMIC)

censoring	count
exact	4,742
left	1,749
right	18

Cells per species

species	cells
<i>E. coli</i>	2,089
<i>S. aureus</i>	1,817
<i>P. aeruginosa</i>	1,223
<i>K. pneumoniae</i>	593
<i>A. baumannii</i>	432
<i>E. faecalis</i>	355

4. Data that was not used

excluded	why
GRAMPA MIC values (51,345 rows)	No medium column exists, so they cannot be reconciled with an MHB-controlled target. GRAMPA contributes sequences only.
Hemolysis data (dbaasp-hc.csv, 34,378 rows / 12,801 peptides with concentration)	Unused.
The 4 smallest species (<i>S. enterica</i> ×2, <i>E. faecium</i> , <i>E. cloacae</i>)	Below ~300 sequences each, hence dropped.

5. Reproducing

uv sync

Corpus (unzip the DBAASP archives first to reproduce)

```

python scripts/dbaasp/02-build-peptides-csv.py
python scripts/dbaasp/04-build-peptide-activity-hc-df.py
python scripts/clean_and_combine/prepare-finetune.py --dbaasp ... --grampa
↳ ... --dramp ... --out ...
python scripts/clean_and_combine/prepare-pretrain.py --marlys ... --out ...

# Cluster peptide sequences
python scripts/cluster_and_seggregate/cdhit-split.py --clstr ... --reference
↳ ... --out ...
python scripts/cluster_and_seggregate/filter-pretrain.py --cluster ...
↳ --pretrain ... --finetune ... --out ...
python -m src.data.clean_corpus
python -m src.data.build_ld_labels

# MIC labels
python scripts/dbaasp/05-standardize-mic.py
python scripts/mic/01-build-mic-labels.py
python tests/test_censoring_sign.py # optional

```